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Name (Print): ____________________

Calculus I (MATH 2200)
Spring 2020
Wednesday, 05/06/2020
Quiz 14: Section 5.4, 6.1

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Section 5.4

3 pts

1. A hiking trail has an elevation given by

$$f(x) = 60x^3 - 650x^2 + 1200x + 4500,$$

where f is measured in feet above sea level and x represents horizontal distance along the trail in miles, with $0 \leq x \leq 5$. Approximately, what is the average elevation of the trail?

If one graphs the function, we may notice that the trail ranges between elevations of about 2000 and 5000 ft. If we let the endpoints of the trail correspond to the horizontal distances $a = 0$ and $b = 5$, the average elevation of the trail in feet is

$$\begin{aligned}\overline{f} &= \frac{1}{b-a}\int_a^b f(x)\ dx \\ &= \frac{1}{5-0}\int_0^5 \left(60x^3 - 650x^2 + 1200x + 4500\right)\ dx \\ &= \frac{1}{5}\left(\frac{60}{4}x^4 - \frac{650}{3}x^3 + \frac{1200}{2}x^2 + 4500x\right)\Bigg|_0^5 \\ &= \frac{11875}{3} \approx 3958.33\end{aligned}$$

So, the average elevation of the trail is slightly less than 3960 ft.

$\boxed{1 \text{ pts}}$ 2. Find the point(s) on the interval $[0, 1]$ at which $f(x) = 2x$ equals its average value on $[0, 1]$.

The average value of f on $[0, 1]$ is

$$\overline{f} = \frac{1}{1-0}\int_0^1 2x \ dx = x^2\Big|_0^1 = 1.$$

Now, we must find the points on $[0, 1]$ at which $f(x) = 1$. Notice, this is equivalent to solving $f(x) - 1 = 0$, i.e. solving the equation $2x - 1 = 0$. Solving, we produce the solution

$$x = \frac{1}{2}.$$

So, for this values of x, we may conclude that f equals its average value on $[0, 1]$.

Section 6.1

$\boxed{3 \text{ pts}}$ 1. A culture of cells in a lab has a population of 100 cells when nutrients are added at time $t = 0$. Suppose the population $N(t)$ (in cells/hr) increases at a rate given by $N'(t) = 90e^{-0.1t}$. Find $N(t)$, for $t \geq 0$.

Knowing that the initial population is $N(0) = 100$ cells, we can find the population $N(t)$ at any future time $t \geq 0$ as follows:

$$\begin{aligned} N(t) &= N(0) + \int_0^t N'(t)\ dt \\ &= 100 + \int_0^t 90e^{-0.1x}\ dx \\ &= 100 + \left(\frac{-90}{0.1} e^{-0.1x} \right) \Big|_0^t \\ &= 100 - 900e^{-0.1t} + 900 \\ &= 1000 - 900e^{-0.1t} \end{aligned}$$

3 pts 2. An artillery shell is fired directly upward with an initial velocity of 300 m/s from a point 30 m above the ground. Assume that only the force of gravity acts on the shell and it produces an acceleration of 9.8 m/s^2. Find the velocity, $v(t)$, of the shell while it is in the air.

We let the positive direction be upward with the origin ($s = 0$) corresponding to the ground. The initial velocity of the shell is $v(0) = 300$ m/s. The acceleration due to gravity is downward; therefore, $a(t) = -9.8\ \text{m/s}^2$. Integrating the acceleration, the velocity is

$$v(t) = v(0) + \int_0^t a(x)\ dx = 300 + \int_0^t -9.8\ dx = 300 - 9.8t.$$